

Tester (QA) Week2 Task - Automated Testing, API Testing

Sunday, September 13, 2026 21:20

Course

Selenium Essential Training

Learning Test Automation

API Testing Foundations

JMETER, swagger, Postman , SMART BEAR tools

Practice: Automate 5 of your Week 1 manual test cases using Selenium (Java or Python). Separately, use Postman to write 5 API test requests (GET/POST) against a free public API (e.g., reqres.in), asserting status codes and response fields.

Setup

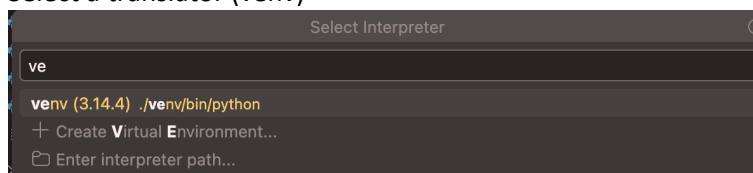
```
(venv) iammin@iammindeMacBook-Air HelloWorld % python3 -m venv venv
(venv) iammin@iammindeMacBook-Air HelloWorld % source venv/bin/activate
(venv) iammin@iammindeMacBook-Air HelloWorld % pip install selenium pytest
```

- **python3 -m venv venv (Create a virtual environment):** This command generates a self-contained directory named venv that holds a local, isolated copy of Python and pip for your current project.
- **source venv/bin/activate (Activate the virtual environment):** This command switches your terminal's active context from the system-wide Python to your isolated local environment. You will see an indicator like (venv) at the beginning of your terminal prompt confirming it is active.
- **pip install selenium pytest (Install dependencies):** This safely downloads and installs the required testing libraries specifically inside your active virtual environment, rather than modifying your Mac's global settings.

Why do we need to do this?

- **System Protection (OS Security):** Modern operating systems like macOS restrict you from installing third-party packages directly into the global Python environment (a security measure known as PEP 668). This prevents users from accidentally breaking core operating system tools that rely on specific system-level Python libraries.
- **Dependency Isolation:** In enterprise software development and QA, different projects require different tool versions. For example, Project A might require an older version of Selenium, while Project B requires the newest release. A virtual environment acts as an isolated sandbox, ensuring the packages and versions you install for one project never conflict with another.

Select a translator (venv)



Code (With the assistant of Claude AI)

See test_youtube_search.py

Run the following command in Terminal:

pytest test_youtube_search.py -v

Result (Not successful)

```
platform darwin -- Python 3.14.4, pytest-9.1.1, pluggy-1.6.0 -- /Users/iammin/Documents/IT/25S1 HIT137 Software Now (Python)/HelloWorld/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/iammin/Documents/IT/25S1 HIT137 Software Now (Python)/HelloWorld
collected 5 items

test_youtube_search.py::test_search_valid_keyword PASSED [ 20%]
test_youtube_search.py::test_search_via_icon FAILED [ 40%]
test_youtube_search.py::test_clear_input_via_x PASSED [ 60%]
test_youtube_search.py::test_empty_search_submission FAILED [ 80%]
test_youtube_search.py::test_minimum_character_autosuggest FAILED [100%]
```

Result (Successful)

```
(venv) iammin@iammindeMacBook-Air HelloWorld % pytest test_youtube_search.py -v
===== test session starts =====
platform darwin -- Python 3.14.4, pytest-9.1.1, pluggy-1.6.0 -- /Users/iammin/Documents/IT/25S1 HIT137 Software Now (Python)/HelloWorld/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/iammin/Documents/IT/25S1 HIT137 Software Now (Python)/HelloWorld
collected 5 items

test_youtube_search.py::test_search_valid_keyword PASSED [ 20%]
test_youtube_search.py::test_search_via_icon PASSED [ 40%]
test_youtube_search.py::test_clear_input_via_x PASSED [ 60%]
test_youtube_search.py::test_empty_search_submission PASSED [ 80%]
test_youtube_search.py::test_special_character_search PASSED [100%]

===== 5 passed in 39.26s =====
o (venv) iammin@iammindeMacBook-Air HelloWorld %
```

Task 2

What is API?

What is an API? In the simplest terms, an API (Application Programming Interface) is like a **waiter** in a restaurant.

When you go to a restaurant, you (the client/app) want to order food, but you can't just walk into the kitchen (the server/database) and cook it yourself. Instead, you call the waiter (the API) and give them your order (send a request). The waiter takes your request to the kitchen, and once the food is ready, they bring it back to your table (return the data). You don't need to know how the kitchen cooks the food; you just need to know how to talk to the waiter.

Here are some free public APIs: <https://www.freepublicapis.com/>

Example:

<https://pokeapi.co/>

The RESTful Pokémon API

Serving over 50 billion API calls each month!

All the Pokémon data you'll ever need in one place,
easily accessible through a modern free open-source RESTful API.

What is the role of PokéAPI in this context?

Imagine you are building a Pokédex app. You need to display Pikachu's height, weight, type, and abilities.

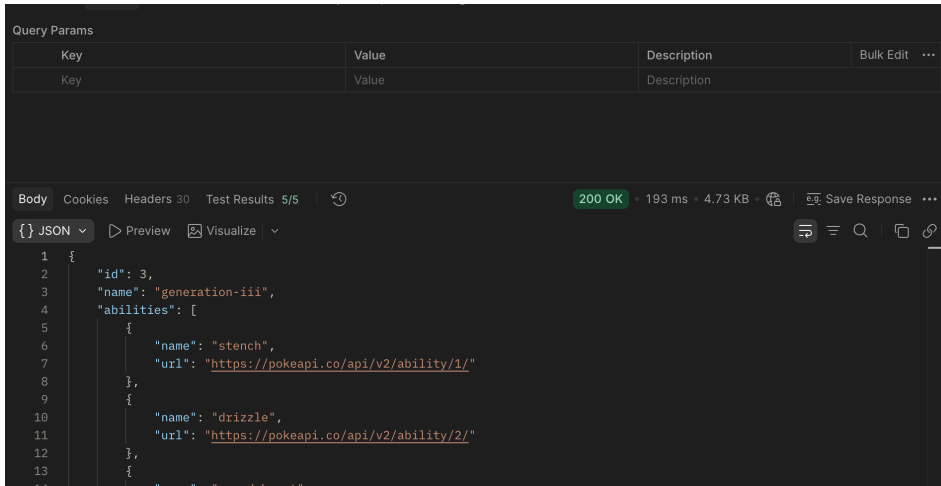
- **Without an API:** You would have to manually research and type out the stats for hundreds of Pokémon into your own database, which would take forever.
- **With PokéAPI:** Your app simply asks the API for the information. Your code sends a quick message (like GET <https://pokeapi.co/api/v2/pokemon/pikachu> (<https://pokeapi.co/api/v2/pokemon/pikachu>)). PokéAPI receives the request, goes into its massive database, fetches Pikachu's exact stats, and hands the data back to your app in a neatly organized format (usually JSON).

Summary: PokéAPI acts as a **messenger that fetches data for you**. It provides a "window" into a massive, already-built database of Pokémon, allowing you to easily pull that information into your own projects without having to collect or store the data yourself.

Step 1: Configure and Execute the API Request

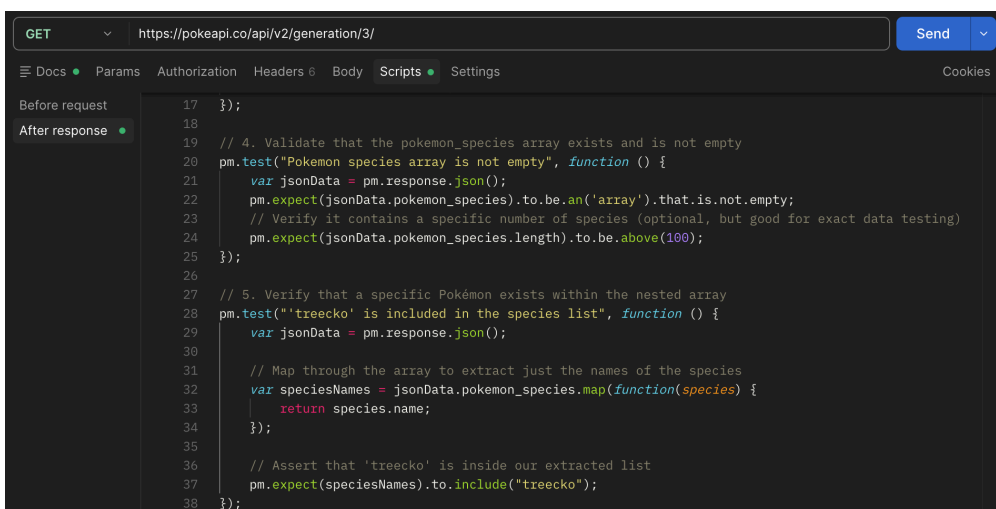
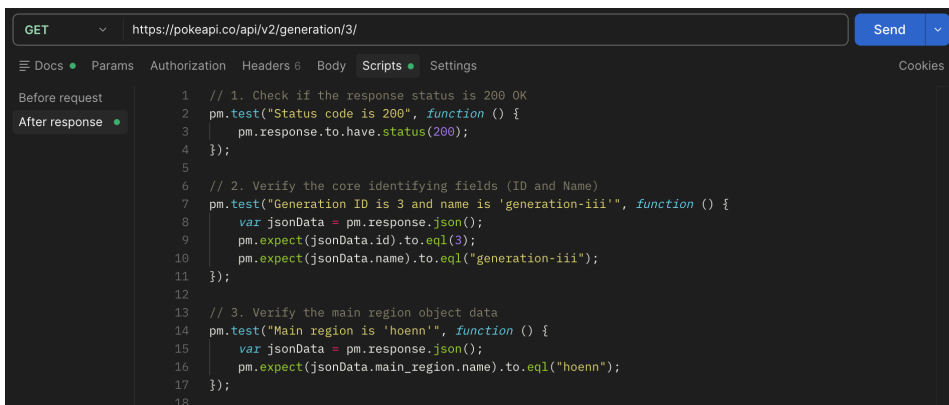
- Select the **GET** method from the dropdown menu.
- Enter the target API endpoint URL (e.g., <https://pokeapi.co/api/v2/generation/3/> (<https://pokeapi.co/api/v2/generation/3/>)) into the address bar.
- Click **Send** to verify the server successfully returns the raw JSON payload in the lower response body.





Step 2: Implement Automated Test Scripts

- Navigate to the **Scripts** tab and select the **After response** section.
- Input the JavaScript assertions to automatically validate the HTTP status code, identify specific core fields, and verify nested array data.



Step 3: Run Tests and Analyze Results

Click Send again to trigger the API request alongside your newly written code.

Open the Test Results tab in the lower response panel to confirm all five assertions successfully passed.





```
PASSED Generation ID is 3 and name is 'generation-iii'
PASSED Main region is 'hoenn'
PASSED Pokemon species array is not empty
PASSED 'treecko' is included in the species list
```

Why We Use Code to Test APIs

- **Eliminates Human Error:** When you request Generation III data from the PokéAPI, the server returns hundreds of lines of raw JSON text. Manually scrolling through that wall of text just to check if "treecko" is in the list, or if the main region is exactly "hoenn", is exhausting and prone to visual mistakes. A coded assertion checks the exact path (`jsonData.main_region.name`) with 100% precision in a fraction of a millisecond.
- **Enables Regression Testing:** In professional software development, you do not test an API just once. Every time a developer updates the backend code, the API must be re-tested to ensure the new changes didn't accidentally break existing features. By writing code, you transform a manual task into a reusable automated script. You can click "Run" and execute hundreds of tests instantly, saving hours of repetitive work.
- **Continuous Integration (CI/CD):** Code-based tests can be run by machines without human intervention. When integrated into a deployment pipeline, your automated Postman scripts act as a security guard. If a developer submits broken code, your script will instantly detect the failure and block the buggy software from being released to the live app.

What It Means When All Test Cases "PASS"

- **The API Contract is Honored:** It proves the server is communicating exactly as promised. It confirms the server is healthy (Status 200), responding with the correct data structures (like arrays instead of plain text), and delivering the exact specific values required by the database.
- **Front-End Safety:** It guarantees that the website or mobile app relying on this API will function correctly and will not crash. For example, if your Pokédex app's user interface is built to loop through the `pokemon_species` array to display a grid of pictures, a "PASS" ensures that array actually exists and is not empty.
- **Deployment Readiness:** It provides the development team with data-driven confidence. A clean pass across all functional, boundary, and negative test cases signals that the core backend logic is stable and ready for production use.